

Fashion-MNIST Classification

Technical Report

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1. Project Overview

The aim of this project was to build a neural network that can classify Fashion-MNIST images into ten clothing categories. I started with a simple baseline and then changed one main factor at a time. This made it easier to see whether each change actually improved the model or only made it more complicated.

2. Data Preparation

Fashion-MNIST was loaded directly from TensorFlow. The original training data was divided into training and validation sets, using a stratified 90/10 split: 54,000 images for training and 6,000 for validation. The official 10,000-image test set was kept separate until the final evaluation. Pixel values were converted to float32 and divided by 255, changing the range from 0-255 to 0-1. Each class contributed 5,400 training images, so class weighting was unnecessary.

3. Model Design

The baseline contains a Flatten layer, one Dense layer with 128 ReLU units, and a 10-unit Softmax output. ReLU allows the model to learn non-linear patterns, while Softmax gives a probability for each class. Sparse categorical cross-entropy was used because the labels are integer class numbers. All models used batch size 128 and a maximum of 30 epochs. Early stopping restored the lowest-validation-loss weights, while ReduceLROnPlateau lowered the learning rate when validation loss stopped improving.

4. Controlled Experiments

- **Experiment 1:** Replaced the 128-unit layer with 256 and 128-unit hidden layers to increase model capacity.
- **Experiment 2:** Kept the 256-128 architecture and added Batch Normalization with 20% Dropout.
- **Experiment 3:** Kept Experiment 2 unchanged and lowered Adam's initial learning rate from 0.001 to 0.0005.

5. Results

The table below summarizes the recorded training and validation results. The model was selected using validation results, not test accuracy.

Model	Params	Run	Best	Time	Train Acc.	Val. Acc.	Val. Loss	Gap
Experiment 2	236,298	24	20	121.1s	0.9480	0.9092	0.2642	0.0388
Experiment 1	235,146	28	24	109.9s	0.9492	0.9063	0.2657	0.0429
Baseline	101,770	30	28	78.8s	0.9369	0.9030	0.2766	0.0339
Experiment 3	236,298	15	11	73.8s	0.9168	0.9028	0.2657	0.0140

Experiment 2 produced the best validation accuracy and loss. Experiment 3 stopped earlier and had the smallest gap, but its validation accuracy was lower, so the learning-rate change was not selected.

6. Final Model Performance

Experiment 2 was selected using validation evidence only. Its restored epoch-20 weights achieved 90.92% validation accuracy and 0.2642 validation loss. On the untouched test set, it achieved 89.76% accuracy, 89.85% macro precision, 89.76% macro recall, and 89.79% macro F1-score. The model exceeded the required 88% test-accuracy target by 1.76 percentage points.

7. Error Analysis

Five incorrect predictions were reviewed in the notebook. Shirt was the most difficult class, with an F1-score of 72.01%. The largest confusions were T-shirt/top predicted as Shirt (132 images), Shirt predicted as T-shirt/top (102 images), and Pullover predicted as Shirt (81 images).

These mistakes are reasonable because several upper-body categories have similar outlines, and Fashion-MNIST images are only 28x28 pixels. Details such as sleeves, collars, and fabric shape are sometimes difficult to see.

8. Overfitting and Generalization

At the selected epoch, Experiment 2 reached 94.80% training accuracy and 90.92% validation accuracy, a 3.88-point gap. This shows some overfitting, but regularization improved it: Experiment 1's gap was 4.29 points. Batch Normalization, Dropout, early stopping, and learning-rate reduction helped control the gap and retain the best validation weights.

9. Conclusion

Increasing capacity improved validation performance but also increased overfitting. Adding Batch Normalization and 20% Dropout produced the strongest result, so Experiment 2 was chosen as the final model. Its 89.76% test accuracy successfully exceeded the target. A compact convolutional neural network would be the next useful experiment because it can preserve spatial relationships and may distinguish similar upper-body garments more effectively.